

Konstantinos Christopher Tsiolis

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PROFILE

I am a third-year PhD student in Statistics at the University of Toronto and the Vector Institute, supervised by Prof. Murat Erdogdu. My research is on theoretical aspects of machine learning, with a particular emphasis on feature learning and scaling laws.

EDUCATION

Doctor of Philosophy (PhD), Statistics

2023 – Present

University of Toronto

Toronto, ON

- Supervisor: Prof. Murat Erdogdu
- Cumulative GPA: 4.0/4.0
- Investigated scaling laws in random features regression with spiked covariance [\[Summary\]](#)
- Theoretical paper on the effect of learning rate on SGD training of two-layer neural networks [\[Paper\]](#)

Master of Science (MSc), Mathematics and Statistics

2022 – 2023

McGill University

Montreal, QC

- Supervisor: Prof. Adam Oberman
- Cumulative GPA: 4.0/4.0
- Wrote a thesis on the link between contrastive learning and similarity kernels [\[Summary\]](#)

Bachelor of Science (BSc), Mathematics and Computer Science (Honours)

2018 – 2021

McGill University

Montreal, QC

- Cumulative GPA: 3.99/4.0
- Conducted research under Prof. Jackie Chi Kit Cheung (2019-2020) and Adam Oberman (2021)
- Implemented and documented the training word embeddings with various context window sizes [\[Summary\]](#)
- Wrote a paper on ensembling pre-trained embeddings from various datasets and algorithms [\[Paper\]](#)
- Developed, implemented, and tested a novel approach for knowledge distillation of deep convolutional networks pre-trained via contrastive learning [\[Code\]](#)

PUBLICATIONS (* DENOTES EQUAL CONTRIBUTION)

- **Konstantinos Christopher Tsiolis**, Alireza Mousavi-Hosseini, and Murat A. Erdogdu. From Information to Generative Exponent: Learning Rate Induces Phase Transitions in SGD. In *Proceedings of the 39th International Conference on Neural Information Processing Systems* (NeurIPS 2025). [\[Paper\]](#) [\[Code\]](#)
- Jingyi He*, **KC Tsiolis***, Kian Kenyon-Dean*, and Jackie Chi Kit Cheung. Learning efficient task-specific meta-embeddings with word prisms. In *Proceedings of the 28th International Conference on Computational Linguistics*, pages 1229–1241. International Committee on Computational Linguistics, 2020. [\[Paper\]](#) [\[Code\]](#)

SKILLS

Programming Languages: Python, R, Java, Bash, C, MATLAB, JavaScript, OCaml

Libraries: PyTorch, NumPy, JAX, scikit-learn, pandas, Matplotlib, tidyverse

Tools: Git, LaTeX, Jupyter, VS Code

AWARDS

NSERC Postgraduate Scholarship – Doctoral (PGS D) (120,000 CAD) 2024 – 2027

Faculty of Arts and Science Top (FAST) Doctoral Fellowship (20,000 CAD) 2023 – 2027

Department of Statistical Sciences Student Leadership Award (1,000 CAD) 2025

Queen Elizabeth II Graduate Scholarship in Science and Technology (QEII-GSST) (15,000 CAD) 2023

FRQNT Master's (B1X) Research Scholarship (17,500 CAD) 2022

- Application ranked 1st out of 17 in Mathematics category

NSERC Canada Graduate Scholarships - Master's Program (CGS M) (17,500 CAD)	2022 – 2023
IVADO Undergraduate Introduction to Research Scholarship (6,500 CAD)	2021
NSERC Undergraduate Student Research Award (4,500 CAD)	2020
McGill University Undergraduate Awards	2018 – 2021
• John Mahon Scholarship (12,000 CAD) • Alan Shaver Scholarship (2,000 CAD) • Anne Molson Scholarship (1,200 CAD) • Emily Ross Crawford Scholarship (1,000 CAD) • Wing Hing Chan Scholarship in Science (500 CAD)	

LEADERSHIP AND SERVICE

Conference Organizer and Reviewer

- Member of the Local Organizing Committee for *The Fast and Curious 2: MCMC in Action* (2025)
- Reviewer for the *Workshop on High Dimensional Learning Dynamics* (2023 – 2025)

President of the Statistics Graduate Student Union (SGSU)

2024 – 2025

University of Toronto Department of Statistical Sciences

Toronto, ON

- Elected representative of the PhD and MSc students in my department
- Organizing academic and social events throughout the year (e.g., Graduate Student Research Day, NSERC Award Application Workshop, Park Day, Game Nights)

Academic Lecturer at the McGill Artificial Intelligence Society

2020 – 2021

McGill University

Montreal, QC

- Volunteer role as part of a student club
- Giving lectures and writing a textbook for a student-run machine learning course
- Holding office hours and one-on-one meetings with students

TEACHING EXPERIENCE

Teaching Assistant

- *At the University of Toronto (2023 – Present)*: STA2311 (Advanced Computational Methods for Statistics I), STA314 (Statistical Methods for Machine Learning I), STA261 (Probability and Statistics II), STA221 (The Practice of Statistics II), STA130 (An Introduction to Statistical Reasoning and Data Science)
- *At McGill University (2021 – 2022)*: MATH 222 (Calculus 3), COMP 251 (Algorithms and Data Structures)
- Prepared material for and conducted lectures, tutorials, exam review sessions, office hours, answering questions on online forums (e.g., Piazza), grading assignments and exams, and invigilation